INCH-POUND
MIL-PRF-23269G
30 November 2007
SUPERSEDING
MIL-PRF-23269F
22 August 2003

PERFORMANCE SPECIFICATION

CAPACITORS, FIXED, GLASS DIELECTRIC, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for established reliability, glass dielectric, fixed capacitors. Capacitors covered by this specification have failure rate levels (FRL) established in accordance with MIL-STD-690. These FRL are established at a 90 percent confidence level and maintained at a 10 percent producer's risk and are based on life tests performed with rated voltage applied at 125°C. An acceleration factor of 5:1 has been used to relate life test data obtained at 150 percent of rated voltage at 125°C to rated voltage at 125°C. A part per million (ppm) quality system is used for documenting and reporting the average outgoing quality of capacitors supplied to this specification. Statistical process control (SPC) techniques are required in manufacturing process to minimize variation in production of ER capacitors supplied to the requirements of this specification.

1.2 Classification. Capacitors covered by this specification are classified by the style, as specified (see 3.1).

* 1.2.1 Part or Identifying Number (PIN). Capacitors covered by this specification are identified by a PIN as shown in the following example:

M23269	/01	-	3001
Performance specification number	Specification slash sheet number		Nonsignificant dash number (see 3.1)

* Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center, Columbus, DSCC-VAT, Post Office Box 3990, Columbus, OH 43218-3990 or e-mailed to capacitorfilter@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-23269/1	-	Capacitors, Fixed, Glass Dielectric, (Axial Wire-Lead Terminals), Established Reliability, Style CYR10.
MIL-PRF-23269/2	-	Capacitors, Fixed, Glass Dielectric, (Axial Wire-Lead Terminals), Established Reliability, Style CYR15.
MIL-PRF-23269/10	-	Capacitors, Fixed, Glass Dielectric, (Radial or Wire-Lead Terminals), Established Reliability, Styles CYR51, CYR52, and CYR53.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Test Methods Standard Electronic and Electrical Component Parts.
MIL-STD-690	-	Failure Rate Sampling Plans and Procedures.
MIL-STD-790	-	Standard Practice for Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications.
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-1276	-	Leads for Electronic Component Parts.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-554-1	-	Assessment of Average Outgoing Quality Nonconforming Levels in Parts Per Million (PPM). (DoD adopted).
EIA-557	-	Statistical Process Control Systems (SPC). (DoD adopted).

(Copies of these documents are available from http://global.ihs.com/ or Global Engineering Documents, Attn: Customer Service Department, 15 Inverness Way East, Englewood CO 80112-5776.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Capacitors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products List (QPL) at the time of award of contract (see 4.4 and 6.3). Authorized distributors which are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL.

3.3 QPL system. The manufacturer shall establish and maintain a QPL system for parts covered by this specification. Requirements for this system are specified in MIL-STD-690 and MIL-STD-790. In addition the manufacturer shall also establish a SPC and PPM system that meets the requirements as detailed in 3.3.1 and 3.3.2 respectively. Reliability of capacitors furnished under this specification shall be established and maintained in accordance with the procedures and requirements specified in MIL-STD-690 and MIL-STD-790 with details specified in 4.4.4.1, and 4.5. The reliability rating is identified by the following FR level symbols:

TABLE I. FRL (established at a 90 percent confidence level).

Symbol	FRL
	Percent/1,000 hour
M	1.0
P	0.1
R	0.01
S	0.001

3.3.1 SPC. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a SPC system that meets the requirements of EIA-557. The manufacturer shall demonstrate control of the temperature coefficient of capacitance and lead integrity in the process.

3.3.2 PPM system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a PPM system of assessing the average outgoing quality of lots in accordance with EIA-554-1. Data exclusion, in accordance with EIA-554-1, may be used with approval of the qualifying activity. The ppm system shall identify the ppm rate at the end of each month and shall be based on six month moving average. Style reporting may include ER style combinations.

3.4 Materials. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Solder. Solder for electrical connections shall be selected to meet the requirements of this specification. In no case shall the solder start to melt at a temperature of less than 200°C.

3.4.2 Soldering flux. The use of materials that leave potentially corrosive residues should be avoided. Where such materials must be used to ensure the satisfactory performance of a process, all potentially corrosive residues must be neutralized and removed to prevent deterioration of the capacitor from the occurrence of related corrosion in application.

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3.5 Interface and physical dimension requirements. Capacitors shall meet the interface and physical dimensions specified (see 3.1).

3.5.1 Terminals.

3.5.1.1 Lead terminals. Lead terminals shall be C, D, N, or W in accordance with MIL-STD-1276. Final finish is the option of the manufacturer.

3.5.2 Solder dip (retinning leads). Only the manufacturer (or their authorized category B distributor) may solder dip/retin the leads of the product supplied to this specification provided the solder dip process (see appendix A) has been approved by the qualifying activity.

* 3.5.2.1 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of capacitor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.5).

3.5.3 Case. Each capacitor shall be enclosed in a suitable case which will protect the capacitor element against the entry of contaminants.

3.5.4 Connections. Electrical connections shall not depend upon the terminals being clamped between a metallic member and an insulating material other than the glass material.

3.6 Seal.

3.6.1 Test I (for transparent cases). When capacitors are tested as specified in 4.7.2.1, there shall be no evidence of dye extending into the active element.

3.6.2 Test II (for opaque cases). When capacitors are tested as specified in 4.7.2.2, the insulation resistance shall be not less than the value specified (see 3.1).

3.7 High voltage stabilization. When capacitors are tested as specified in 4.7.3, there shall be no evidence of damage, arcing, or breakdown.

3.8 Insulation resistance. When measured as specified in 4.7.4, the insulation resistance shall be not less than the value specified (see 3.1).

3.9 Capacitance. When measured as specified in 4.7.5, the capacitance shall be within the applicable tolerance of the nominal value specified (see 3.1).

3.10 Dissipation factor. When measured as specified in 4.7.6, the dissipation factor shall not exceed the value specified (see 3.1).

* 3.11 Thermal shock. When tested as specified in 4.7.7, there shall be no short-circuiting and capacitors shall meet the following requirements:

Insulation resistance:	Shall be not less than 100,000 megohms.
Capacitance:	Shall change not more than 0.5 percent of the nominal value or 0.5 pF, whichever is greater, from the initial value obtained when measured as specified in 4.7.5.
Dissipation factor:	Shall not exceed the value specified.
Visual examination:	There shall be no evidence of corrosion or mechanical damage.

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3.12 Quality factor (Q). When measured as specified in 4.7.8, the Q shall be not less than the value shown on figure 1, unless otherwise specified (see 3.1).

3.13 Shock, specified pulse. When capacitors are tested as specified in 4.7.9, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, open-circuiting or short-circuiting, or evidence of arcing or mechanical damage.

3.14 Vibration, high frequency. When capacitors are tested as specified in 4.7.10, there shall be no intermittent contacts of 0.5 ms or greater duration, open-circuiting or short-circuiting, or evidence of mechanical damage.

3.15 Solderability. When capacitors are tested as specified in 4.7.11, the dipped portion of the terminals shall conform to the solid-wire termination criteria of method 208 of MIL-STD-202.

3.16 Terminal strength. When capacitors are tested as specified in 4.7.12, there shall be no loosening or rupturing of the terminals, and no damage to the terminals or seal.

3.17 Barometric pressure. When capacitors are tested as specified in 4.7.13, there shall be no evidence of damage, arcing, or breakdown.

3.18 Temperature coefficient and capacitance drift. When measured as specified in 4.7.14, the temperature coefficient and capacitance drift shall be as specified (see 3.1).

3.19 Salt atmosphere. When capacitors are tested as specified in 4.7.15, there shall be no harmful corrosion or mechanical damage, and the marking shall remain legible.

NOTE: Harmful corrosion shall be construed as being any type of corrosion which in any way interferes with the mechanical or electrical performance of the capacitor.)

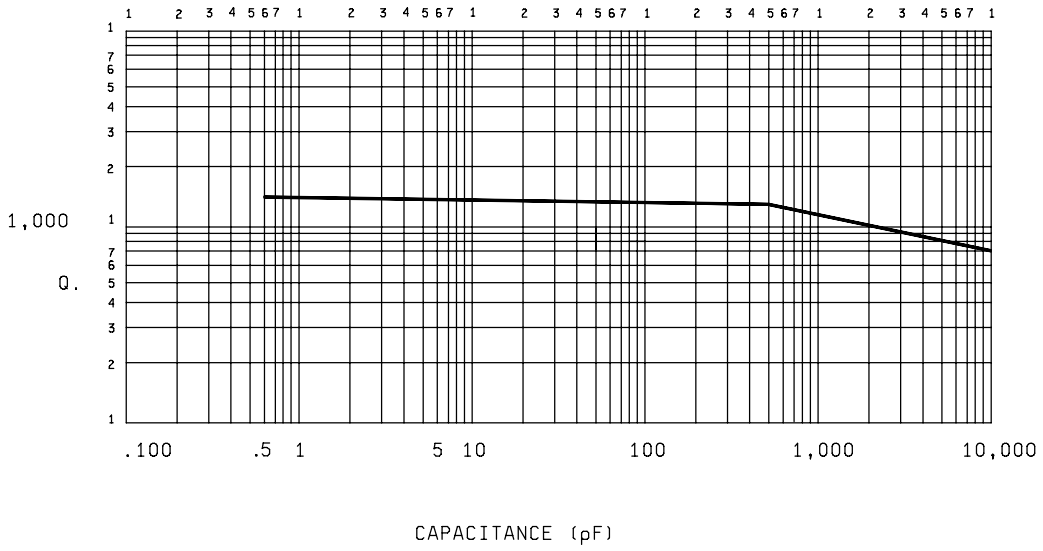


FIGURE 1. Quality factor (Q) at 1 MHz.

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3.20 Fungus. The manufacturer shall certify that all external materials are fungus resistant or shall perform the test specified in 4.7.16. When tested as specified in 4.7.16, there shall be no evidence of fungus growth on the external surface of the capacitor.

3.21 Resistance to solvents. When capacitors are tested as specified in 4.7.17, there shall be no evidence of mechanical damage, and the marking shall remain legible.

Insulation resistance:	Shall be not less than the initial requirement.
* Capacitance:	Shall change not more than 0.5 percent of nominal value or 0.5 pF, whichever is greater, from the initial value obtained when measured as specified in 4.7.5.
Dissipation factor:	Shall not exceed the initial requirement.
Visual examination:	There shall be no evidence of mechanical damage.

3.22 Resistance to soldering heat. When tested as specified in 4.7.18, capacitors shall meet the following requirements:

Insulation resistance:	Shall be not less than the initial requirement.
Capacitance:	Shall change not more than 0.5 percent of nominal value or 0.5 pF, whichever is greater, from the initial value obtained when measured as specified in 4.7.5.
Dissipation factor:	Shall not exceed the initial requirement.
Visual examination:	There shall be no evidence of mechanical damage

3.23 Moisture resistance. When tested as specified in 4.7.19, capacitors shall meet the following requirements:

Visual examination:	No mechanical damage. Marking shall remain legible.
Insulation resistance:	Unless otherwise specified (see 3.1), not less than 10 percent of the initial 25°C requirement.
Capacitance:	Unless otherwise specified (see 3.1), change not to exceed ±10 percent from initial measured value.

3.24 Life. When tested as specified in 4.7.20, capacitors shall meet the following requirements, as applicable.

3.24.1 2,000 hour (qualification inspection) (see 4.7.20.1).

Insulation resistance:	Shall be not less than the initial requirement.
* Capacitance:	Shall change not more than the percent specified (see 3.1) of nominal value or 0.5 pF, whichever is greater, from the initial value obtained when measured as specified in 4.7.5.
Dissipation factor:	Shall not exceed the value specified (see 3.1).
Visual examination:	There shall be no evidence of mechanical damage.

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3.24.2 6,000 hour (group C inspection) and extended life (see 4.7.20.2 and 4.7.20.3).

Insulation resistance:	Shall be not less than 10,000 megohms.
* Capacitance:	Shall change not more than ±2.0 percent of nominal value or 0.5 pF, whichever is greater, from the initial value obtained when measured as specified in 4.7.5. For FRL calculation, the ±2.0 percent criteria shall govern from 2,000 hours to 30,000 hours.
Dissipation factor:	Shall not exceed 1.0 percent.
Visual examination:	There shall be no evidence of mechanical damage.

* 3.25 Marking. Capacitors shall be permanently marked with the Part or Identifying Number (PIN), “JAN” marking, date code, lot symbol, and manufacturer’s source code in accordance with MIL-STD-1285 (see 3.1). Paper labels shall not be used. Each capacitor shall be legibly marked using smear-resistance ink which will withstand the environmental conditions specified herein. Marking shall remain legible after all tests. Manufacturing records shall include these same date codes and lot symbols. The date code, lot symbol, source code, capacitance, capacitance tolerance, and rated voltage shall be marked on the unit package. The nominal capacitance value, expressed in pF, is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. When fractional values of a pF or values of less than 10 pF are required, the letter “R” shall be used to indicate the decimal point and the succeeding digits or the group shall represent significant figures. Example: 1R5 indicates 1.5 pF.

3.25.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to and meeting all of the criteria specified herein and in applicable specification, shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated specifications, the manufacturer shall remove completely the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 1,586,261 for the certification mark "J".

3.25.2 Substitution of FRL. A manufacturer may supply to all higher FRL than to which they are qualified. Items of an exponential FRL as shown in table II and marked to lower FRL, with acquiring agency approval, are substitutable for higher FRL, and shall not be remarked unless specified in the contract or order (see 6.2), the lot date codes on the parts are unchanged, and the workmanship criteria is met.

TABLE II. FRL substitutability.

Parts qualified to FRL	Are substitutable for FRL
S	M, P, and R
R	M and P
P	M

3.25.3 Substitutability of capacitance tolerance and rated voltages. Parts qualified and marked to tighter capacitance tolerance or higher rated voltage, with acquiring agency approval, are substitutable for parts marked to looser capacitance tolerance or lower rated voltage, provided all other values, such as case size, characteristic, and leads are the same. The substitutable parts shall not be remarked unless specified in the contract or order (see 6.2). In the event the capacitance tolerance or voltage rating is remarked, the lot date codes on the parts shall not be changed and the workmanship criteria shall be met.

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3.25.4 Full marking. Unless otherwise specified (see 3.1), capacitors shall be marked with the "JAN" or "J" marking, PIN (see 1.2.1), date code and lot number, manufacturer's name (not trademark) or the Commercial and Government Entity (CAGE) code, voltage, capacitance, and capacitance tolerance. There shall be no space between the symbols which comprise the PIN. The date code and lot number shall consist of the year, week, and lot code. For example: The third week of 1984 would be 8403.

3.26 Recycled recovered or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.27 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects which will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Conformance inspection (see 4.6).

4.2 QPL system. The manufacturer shall establish and maintain a QPL system as described in 3.3. Evidence of such compliance is a prerequisite for qualification and retention of qualification.

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.1 Accuracy of test voltage measurements. Accuracy of all test voltage measurements shall be ±2.0 percent of the specified voltage.

4.3.2 Referenced measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurements shall be considered the last measurement made at 25°C ±3°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2.0 percent or less of the specified test voltage.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. The decision as to whether or not the product is to be included on the QPL shall be made at the conclusion of the 2,000-hour life test.

4.4.1 Sample size. The number of capacitors to be subjected to qualification inspection shall be as specified in Appendix A of this specification.

4.4.2 Inspection routine. The sample shall be subjected to the inspection specified in table III, in the order shown. All sample units shall be subjected to the inspection of group I and group II. The sample units shall then be divided as specified in table III for group III to group VI inclusive, and subjected to the inspections for their particular groups; one group shall be subjected to the accelerated condition and the other group to the rated condition (see 4.4.4.1a).

4.4.3 Failures. Failures in excess of those allowed in table III shall be cause for refusal to grant qualification approval.

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TABLE III. Qualification inspection.

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected		Number of failures allowed 1/
Group I					
Seal	3.6	4.7.2	115		
High voltage stabilization	3.7	4.7.3			
Group II					
Visual and mechanical examination (external) 2/ 3/	3.1, 3.4, 3.5, 3.25, and 3.27	4.7.1	4/ 115		1
Insulation resistance (at 125°C and 25°C)	3.8	4.7.4			
Capacitance	3.9	4.7.5			
Dissipation factor	3.10	4.7.6			
Group III					
Thermal shock	3.11	4.7.7	12		1
Quality factor (Q)	3.12	4.7.8			
Shock, specified pulse	3.13	4.7.9			
Vibration, high frequency	3.14	4.7.10			
Group IV					
Solderability	3.15	4.7.11	12		1
Terminal strength	3.16	4.7.12			
Barometric pressure	3.17	4.7.13			
Temperature coefficient and capacitance drift	3.18	4.7.14			
Salt atmosphere	3.19	4.7.15			
Group V					
Fungus 5/	3.20	4.7.16	5	10	1
Resistance to solvents	3.21	4.7.17	5		
Resistance to soldering heat	3.22	4.7.18			
Moisture resistance	3.23	4.7.19			
Group VI					
Life (rated condition)	3.24.1	4.7.20.1	50		1
Life (accelerated condition)	3.24.2	4.7.20.3	30		

1/ A specimen having one or more defects shall be considered as a single failure.
2/ Nondestructive tests.
3/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect marking.
4/ One additional sample unit is included in each sample of 115 sample units to permit substitution for the failure allowed in group II.
5/ Certification of fungus resistance may be substituted for testing.

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4.4.4 FRL and quality level verification.

4.4.4.1 FR qualification and lot conformance FR inspection. FR qualification and lot conformance FR inspection shall be in accordance with the general and detailed requirements of MIL-STD-690 with the following details:

a. Procedure I: Qualification at the initial FRL. Level M of FRSP-90 shall apply. Sample units which have been subjected to the qualification inspection specified in group VI, table III (see 4.4.2) shall be continued on test as specified in 4.7.20.3.1.

b. Procedure II: Extension of qualification to lower FRL. To extend qualification to the R and S FRL, data from two or more styles of similar construction (see 4.6.1.1) may be combined.

c. Procedure III: Maintenance of FRL qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification.

d. Procedure VII: Lot conformance FR inspection. Lot conformance FR inspection shall be in accordance with 4.6.1.2.

4.4.4.2 Quality level verification. The contractor is responsible for establishing a quality system to verify the ppm defect level of lots. The ppm defect level shall be based on a 6-month moving average.

4.5 Verification of qualification. Every six months, the manufacturer shall verify the retention of qualification to the qualifying activity. Continuation of qualification shall be based on meeting the following requirements:

a. MIL-STD-790 program.

b. The capacitor design has not been modified.

c. Lot rejection for group A inspection does not exceed 10 percent or one lot, whichever is greater.

d. Periodic group C inspection.

e. FRL's.

f. PPM assessment.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of capacitors from the same production line or lines, of one or more styles, from the same basic design, produced under essentially the same conditions, and offered for inspection during a period not exceeding 1 month. Each lot shall be kept separate from every other lot. The sample selected from the lot shall be representative of the capacitance values and case sizes in the lot. All sample units belonging to a lot shall be identified by means of a code symbol (either letters or numbers, at the option of the manufacturer). The following styles are considered to be of similar construction and may be combined for conformance testing and FRL maintenance:

*

Group		Styles
1	-	CYR10 (/1), CYR15 (/2).
2	-	CYR51, CYR52, and CYR53 (/10).

A separate sample for each of the two groups shall be submitted for conformance testing and FRL maintenance.

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4.6.1.2 Production lot. A production lot shall consist of all capacitors of the same style, voltage rating, nominal capacitance value, voltage temperature characteristic, and termination finish. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.6.1.3 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table IV, in the order shown.

4.6.1.3.1 Subgroup 1 tests. The subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Capacitors failing the tests of subgroup 1 shall be removed from the lot. If, during the 100 percent inspection, screening requires more than 5 percent of the capacitors to be discarded, the entire production lot shall be rejected.

* TABLE IV. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	Sample	Lot criteria
Subgroup 1				
Seal	3.6	4.7.2	100% inspection	
High voltage stabilization	3.7	4.7.3		
Insulation resistance (at 25°C)	3.8	4.7.4		
Capacitance	3.9	4.7.5		
Dissipation factor	3.10	4.7.6		
Thermal shock (styles CYR51, CYR52, and CYR53 only)	3.11	4.7.7		
Subgroup 2 1/				
Visual and mechanical examination				
Materials	3.4	4.7.1	13	0 failures
Physical dimensions	3.1	4.7.1		
Design and construction (other than physical dimensions)	3.5	4.7.1		
Marking 2/	3.25	4.7.1		
Workmanship	3.27	4.7.1		
Subgroup 3 3/				
Solderability	3.15	4.7.11	5	0 failures

* 1/ The manufacturer may request the deletion of the subgroup 2, visual and mechanical examination, provided an in-line or process control system for assessing and assuring the visual and mechanical requirements are met, can be validated and approved by the qualifying activity. Deletion of this examination does not relieve the manufacturer from meeting these requirements.

2/ Marking defects are based on visual inspection only.

* 3/ The manufacturer may request the deletion of the subgroup 3 solderability test, provided an in-line or process control system for assessing and assuring the solderability of leads can be validated and approved by the qualifying activity. Deletion of the test does not relieve the manufacturer from meeting this test requirement.

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4.6.1.3.2 Manufacturer's production inspection. If the manufacturer performs tests equal to or more stringent than those specified in subgroup 1, table IV, as the final step of their production process, group A, subgroup 1 inspection may be waived. Authority to waive the subgroup 1 inspections shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 1.

b. Manufacturer subjects 100 percent of the product supplied under this specification to their production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.6.1.3.3 Rejected lots. Production lots exceeding the 5 percent PDA of group A, subgroup 1 inspection shall be segregated from new lots that have passed inspection. Lots rejected may be offered for acceptance only if the manufacturer 100 percent retests to the requirements of subgroup 1. Resubmitted lots shall be kept separate and shall be clearly identified as resubmitted lots. If, during the 100 percent re-inspection to subgroup 1, the lot exceeds 3 percent defective, the lot shall be rejected and shall not be resubmitted.

4.6.1.3.4 Subgroup 2 tests.

4.6.1.3.5 Subgroup 2 tests. Subgroup 2 tests shall be performed on an inspection lot basis. A sample of 13 parts shall be randomly selected. If one or more defects are found, the lot shall be 100 percent screened for that defect. A new sample shall then be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3.6 Subgroup 3 tests (solderability).

4.6.1.3.6.1 Inspection lot. An inspection lot for the purpose of subgroup 3 (solderability) testing shall consist of all lots manufactured with the same diameter lead wire and offered for inspection within the same work week. Each manufacturing lot shall be kept separate from every other lot. All samples belonging to a manufacturing lot shall be identified to that lot. Means of identification is at the option of the manufacturer.

4.6.1.3.6.2 Sampling plan. Five samples shall be selected randomly from each inspection lot and subjected to the solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

* 4.6.1.3.6.3 Rejected lots. In the event of one or more defects, the inspection lot shall be rejected. The manufacturer may rework the lot as follows: Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.7.11. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure specified in Appendix A of this specification.

4.6.1.3.6.4 Disposition of samples. The solderability test is considered a destructive test, and samples submitted to the solderability test shall not be supplied on the contract.

4.6.1.3.6.5 PPM calculations. The manufacturer shall establish a ppm system in accordance with 3.3.2 for accessing and calculating average outgoing quality of capacitors. A ppm rate combining insulation resistance, capacitance, and dissipation factor, shall be assessed for lots that have passed the group A inspection. The manufacturers ppm system shall also address rectification procedures for lots failing ppm assessment. Data from the rectification process shall not be used to calculate ppm.

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* 4.6.2 Periodic group C inspection. Group C inspection shall be performed on sample units which have been subjected to and have passed the applicable tests for group A inspection and shall consist of the tests specified in table V, in the order shown. Delivery of products which have passed group A inspection shall not be delayed pending the results of periodic inspection.

4.6.2.1.1 Sampling plan.

* 4.6.2.1.1.1 Group I. Fifty-one sample units shall be taken from production every 2 months and subjected to the applicable tests for their particular subgroup. Allowable failures shall be as specified in table V.

4.6.2.1.1.2 Group II. A minimum of 10 sample units shall be selected from each inspection lot produced during a 2-month period. Allowable failures shall be as specified in MIL-STD-690. The accumulated data shall be used for maintenance and extension of FR qualification.

4.6.2.1.2 Noncompliance. If a sample unit fails to pass group C inspection, the manufacturer shall notify the qualifying activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after re-inspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

* TABLE V. Group C inspection.

Test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed	
Group I					
Subgroup IA 1/					
Thermal shock	3.11	4.7.7	24	1	
Subgroup IB 1/					
Quality factor(Q)	3.12	4.7.8			
Shock, specified pulse	3.13	4.7.9	12	1	
Vibration, high frequency	3.14	4.7.10			
Subgroup IC 1/					1
Terminal strength	3.16	4.7.12			
Barometric pressure	3.17	4.7.13	12	1	
Temperature coefficient and capacitance drift	3.18	4.7.14			
Salt atmosphere	3.19	4.7.15			
Subgroup ID 1/					
Resistance to solvents	3.21	4.7.17			
Resistance to soldering heat	3.22	4.7.18	3	0	
Moisture resistance	3.23	4.7.19			
Group II					
Life (accelerated conditions)	3.24.2	4.7.20.2	10 minimum	(See 4.6.2.1.1.2)	

* 1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, then these tests, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform these tests every three years after the deletion as part of long term design verification. If the design, material, construction, or processing of the part is changed or, if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.

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4.7 Methods of inspection.

4.7.1 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.25, and 3.27).

4.7.2 Seal (see 3.6). Capacitors shall be tested in accordance with 4.7.2.1 or 4.7.2.2, as specified (see 3.1).

4.7.2.1 Test I (for transparent cases). Capacitors shall be immersed in a fluorescent penetrant for six and one-half minutes and a pressure of 20 pounds per square inch gauge (psig) shall be maintained for 3 minutes. Upon removal from the penetrant, capacitors shall be cleaned, dried, and inspected under ultra-violet light for penetration of the dye.

4.7.2.2 Test II (for opaque cases). Capacitors shall be subjected to a saturated steam atmosphere of 5 psig for a period of 20 minutes to 30 minutes. Insulation resistance shall be measured as specified in 4.7.4 within 5 minutes after removal from steam atmosphere.

4.7.3 High voltage stabilization (see 3.7). Capacitors rated at 300 volts and above shall be subjected to 1,500 volts ±30 volts dc; capacitors rated at less than 300 volts shall be subjected to 400 percent ±2 percent of the dc rated voltage at room temperature for 50 hours +10 hours, -0 hours. During this test, capacitors shall be adequately protected against temporary voltage surges of 10 percent or more of the test voltage. After the test, capacitors shall show no damage, arcing, or breakdown.

4.7.4 Insulation resistance (see 3.8). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test potential: Shall not exceed the rated voltage (see 3.1).

b. Points of measurement: From terminal to terminal.

4.7.5 Capacitance (see 3.9). Capacitance shall be measured in accordance with method 305 of MIL-STD-202. The following details shall apply:

a. Test frequency: 1 MHz ±50 kHz when the nominal capacitance is 1,000 pF or less, and 1kHz ±50 Hz, when the nominal capacitance is greater than 1,000 pF. At the option of the manufacturer, capacitance measurements may be made at any frequency from 1 kHz to 1 MHz and referred to measurements at 1 MHz and 1 kHz, respectively.

* b. Limit of accuracy: Shall be ±0.2 percent of nominal capacitance value or ±0.2 pF, whichever is greater.

4.7.6 Dissipation factor (see 3.10). Dissipation factor shall be measured at a frequency of 1 kHz ±100 Hz. Measurement accuracy shall be within ±2.0 percent of 0.0005, whichever is greater.

4.7.7 Thermal shock (see 3.11). Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

a. Test condition letter: B

b. Measurements before cycling: Capacitance shall be measured as specified in 4.7.5.

* c. Measurements and examination after cycling: Insulation resistance, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6 respectively. Capacitors shall be visually examined for evidence of mechanical damage.

4.7.8 Quality factor (Q) (see 3.12). Capacitors shall be tested in accordance with method 306 of MIL-STD-202. The test frequency shall be 1 MHz ±50 kHz.

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4.7.9 Shock, specified pulse (see 3.13). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

a. Mounting: Capacitors shall be rigidly mounted by the body to the test apparatus in such a manner that the mounting method does not damage the capacitors.

b. Test condition letter: I (100 g's).

c. Measurements during shock: During shock, an electrical measurement shall be made to determine intermittent contacts or open-circuiting or short-circuiting. The accuracy of the detecting equipment shall be sufficient to detect any interruption of 0.5 ms or greater duration.

After the test, capacitors shall be examined for evidence of arcing and mechanical damage.

4.7.10 Vibration, high frequency (see 3.14). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

a. Mounting: Capacitors shall be rigidly mounted by the body to the test apparatus in such a manner that the mounting method does not damage the capacitors.

b. Test condition letter: D (20 g's).

c. Measurements during shock: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent contacts or open-circuiting or short-circuiting. The accuracy of the detecting equipment shall be sufficient to detect any interruption of 0.5 ms or greater duration.

After the test, capacitors shall be visually examined for evidence of arcing and mechanical damage.

4.7.11 Solderability (see 3.15). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. Number of terminations of each capacitor to be tested: Two.

b. Depth of immersion in flux and solder: Both terminals shall be immersed to within .05 inch of the capacitor body.

4.7.12 Terminals strength (see 3.16). The number of sample units shall be divided into two equal groups; a different group shall be used for each of the following tests:

a. Pull test: Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following detail and exception apply:

(1) Test condition letter A.

(2) Applied force: Four and one-half pounds.

b. Bend test: A 1-pound weight shall be hung from each terminal in turn at a point .750 inch (19.05 mm) from its point of egress from the capacitor body, and the body shall be rotated about an axis which is perpendicular to the axis of the terminals and parallel to the largest face, through 90 degrees, back through 180 degrees, and then return to its original position (see figure 2).

After the test, capacitors shall be visually examined for evidence of loosening or rupturing of the terminals, or damage to the terminals or seal.

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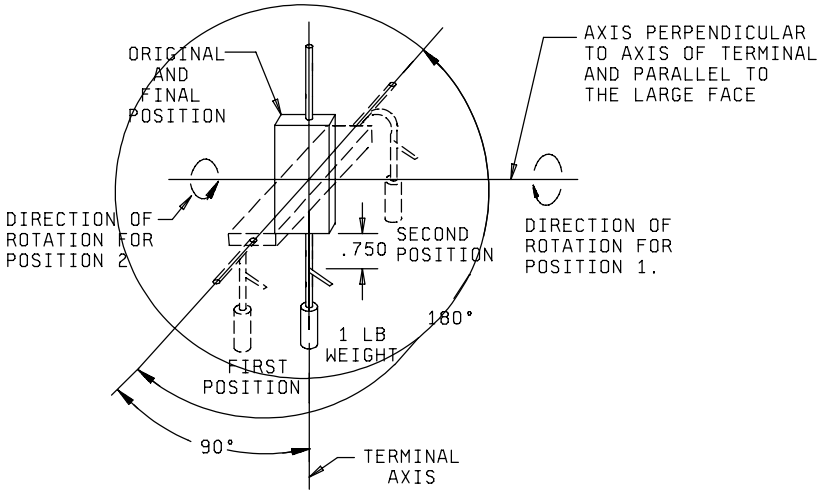


FIGURE 2. Bend test.

4.7.13 Barometric pressure (see 3.17). Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting: Not applicable.

b. Test condition letter: D (100,000 feet).

c. Tests during subjection to reduced pressure: a minimum of 150 percent - 2 percent of rated voltage shall be applied between the terminals for not less than 1 second nor more than 5 seconds.

After the test, capacitors shall be examined for evidence of damage, arcing, and breakdown.

4.7.14 Temperature coefficient and capacitance drift (see 3.18). Capacitance shall be measured as specified in 4.7.5, except that measurements shall be made in the order and at the temperature shown in the following, at a frequency of 100 kHz ±10 kHz. The reference frequency at which measurements are made shall not drift more than ±50 Hz during the test. An accuracy of ±0.025 percent of nominal capacitance +0.05 pF shall be maintained for measurement of capacitance change.

a. For qualification inspection: +25°C, ±2°C; -55°C +0°C, -2°C; -10°C ±2°C; +25°C ±2°C; +65°C ±2°C; +85°C +2°C, -0°C; +125°C +2°C, -0°C; and +25°C ±2°C.

b. For quality conformance inspection: +25°C, ±2°C; -55°C +0°C, -2°C; +25°C ±2°C; +125°C +2°C, -0°C; and +25°C ±2°C, respectively.

The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals at that temperature indicate no change in capacitance. The temperatures at the time of measurement shall be measured at an accuracy of ±1.0 percent of the temperature difference between the nominal test temperature and the nominal reference temperature +0.5 °C.

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4.7.14.1 Temperature coefficient. The temperature coefficient shall be computed as follows:

$$TC = \frac{(C_2 - C_1)\,10^6}{(T_2 - T_1)\,C_1}$$

Where:

TC = Temperature coefficient (in parts per million per degree Celsius).
C_1 = Capacitance (in pF at the middle 25°C (reference temperature)).
C_2 = Capacitance (in pF at test temperature).
T_1 = 25°C.
T_2 = Test temperature (in degrees Celsius).

4.7.14.1.1 Continuous curve temperature coefficient. As an alternate to the measurements specified in 4.7.14, a continuous curve of capacitance versus temperature may be produced by subjecting the capacitors to a slowly varying temperature. The temperature shall be varied from +25°C to -55°C to +125°C to +25°C. A temperature-sensing device shall be embedded in a dummy capacitor in a manner to assure accurate internal readings in the sample under test. Temperature shall be varied slowly enough to produce a smooth uniform curve with no loops at -55°C or +125°C. Other test conditions shall be as specified in 4.7.14.

4.7.14.2 Capacitance drift. Capacitance drift shall be computed by dividing the greatest single difference between any two of the three values recorded at +25°C by the intermediate value recorded at +25°C.

4.7.15 Salt atmosphere (see 3.19). Capacitors shall be tested in accordance with method 101 of MIL-STD-202. The following detail and exception shall apply:

a. Applicable salt solution: Five percent.

b. Test condition: B.

c. Examination after test: Capacitors shall be visually examined for evidence of harmful corrosion and mechanical damage.

4.7.16 Fungus (see 3.20). Capacitors shall be tested in accordance with method 508 of MIL-STD-810. Pretest and post-test measurements are not required. After the test, capacitors shall be visually examined for evidence of fungus growth.

4.7.17 Resistance to solvents (see 3.21). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details and exceptions shall apply:

a. Portion of specimen to be brushed: That portion on which marking is present.

b. Number of specimens to be tested: As specified in applicable inspection tables.

c. Permissible extent of damage: As specified in 3.21.

4.7.18 Resistance to soldering heat (see 3.22). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exception shall apply:

a. Test condition letter: C.

b. Cooling time prior to measurement after test: 10 minutes ±1 minute.

c. Measurements and examination after test: Insulation resistance, capacitance, and dissipation factor shall be measured at 25°C as specified in 4.7.4, 4.7.5, and 4.7.6, respectively. Capacitors shall then be examined for evidence of mechanical damage.

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4.7.19 Moisture resistance (see 3.23). Following resistance to soldering heat, capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Capacitors shall be mounted by their terminals in a manner that will keep the capacitor body from touching the test fixture.

b. Initial measurements: Not applicable.

c. Number of cycles: 20 continuous cycles with step 7a applying for the first 10 cycles only. Step 7b is not applicable.

d. Polarization voltage: For the first ten cycles, a dc potential of 90 volts to 100 volts shall be applied across the capacitor terminals. Once each day, a check shall be made to determine whether a capacitor has shorted.

e. Final measurements: After the final cycle and between one hour and four hours after removal from the chamber, insulation resistance shall be measured at room ambient conditions as specified in 4.7.4. Artificial drying shall not be permitted. Between 4 hours and 24 hours after this measurement, capacitors shall be visually examined for evidence of corrosion or mechanical damage; capacitance, and dissipation factor shall then be measured as specified in 4.7.5 and 4.7.6, respectively.

4.7.20 Life (see 3.24).

4.7.20.1 2,000 hour (qualification inspection) (see 3.24.1). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Distance of temperature measurements from specimens, in inches: Not applicable.

b. Test temperature and tolerance: +125°C +4°C, -0°C.

c. Operating conditions: A minimum of dc rated voltage -2 percent of dc rated voltage shall be applied to the capacitors being tested under rated conditions; a minimum of 150 percent -2 percent of the dc rated voltage shall be applied to the capacitors being tested under accelerated conditions. The maximum surge current shall be not more than 50 mA. A current-limiting resistor shall be inserted into the circuit.

d. Test condition letter: F (2,000 hours +72 hours, -0 hours).

e. Measurements after exposure: At the conclusion of this test, the insulation resistance, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

f. Examination after test: Capacitors shall be visually examined for evidence of corrosion or mechanical damage.

4.7.20.2 6,000 hour (group C inspection) (see 3.24.2). Except as specified in the following, capacitors shall be tested as specified in 4.7.20.1:

a. Test duration: 6,000 hours +96 hours, -0 hours at accelerated condition only.

b. Measurements during and after exposure: Insulation resistance, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively, after 2,000 hours +72 hours, -0 hours and every 2,000 hours +96 hours, -0 hours thereafter until a total of 6,000 hours +96 hours, -0 hours have elapsed.

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4.7.20.3 Extended life (see 3.24.2).

4.7.20.3.1 Following 2,000 hour qualification test. Capacitors tested under accelerated conditions shall be continued on test for an additional 4,000 hours +96 hours, -0 hours; measurements during and after exposure shall be as specified in 4.7.20.2. Capacitors tested under rated conditions shall be continued on test for an additional 28,000 hours +96 hours, -0 hours; measurements during and after exposure shall be accomplished after 2,000 hours +96 hours, -0 hours thereafter up to 10,000 hours; then every 5,000 hours +96 hours, -0 hours until a total of 30,000 hours +96 hours, -0 hours have elapsed.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These capacitors are intended for use in any equipment where known orders of reliability are required, and are primarily designed as a substitute for mica-dielectric capacitors as a step toward conservation of critical mica. They are effective substitutes for mica-dielectric capacitors and can be employed for many applications where mica-dielectric capacitors are used, provided consideration is given to the differences in temperature coefficient and dielectric loss. Capacitors covered by this specification are military unique due to the fact that these devices must be able to operate satisfactorily in military systems under the following demanding environmental conditions: shock, vibration, acceleration, extreme moisture, vacuum, extended life of 30,000 hours and more, and high operating temperatures such as experienced in missile-borne and space electronic equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification, the applicable specification sheet, and the complete PIN (see 3.1).

* b. Remarking of capacitors (see 3.25.2 and 3.25.3).

c. Packaging requirements (see 5.1).

d. Lead finish (see 3.5.1.1).

e. Retinning (hot solder dip), if required (see 3.5.2).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Defense Supply Center, Columbus, ATTN: DSCC-VQP, PO Box 3990, Columbus, OH 43218-3990, or by e-mail to vqp.chief@dla.mil.

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6.4 Subject term (key word) listing.

Parts Per Million (ppm)
Statistical process control (SPC)

* 6.5 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B545, Standard Specification for Electrodeposited Coating of Tin.

* 6.6. Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmentally Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals is available on their website at http://www.epa.gov/epaoswer/hazwaste/minimize/chemlist.htm. Further information is available at the following EPA site: http://www.epa.gov/epaoswer/hazwaste/minimize/. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of the materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see Section 3).

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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APPENDIX A

PROCEDURE FOR QUALIFICATION INSPECTION

A.1. SCOPE

A.1.1 Scope. This appendix details the procedure for submission of samples for qualification inspection of capacitors covered by this specification. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

A.3. SUBMISSION

A.3.1 Sample.

A.3.1.1 Single-type submission. A sample consisting of 115 sample units of each type for which qualification is sought shall be submitted (see A.3.1.2).

A.3.1.2 Combined submission. A sample consisting of 115 sample units of the highest capacitance value shall be submitted for each style for which qualification is sought. For those styles containing two dc voltage ratings, an equal number of sample units (58 each) shall be submitted representing the largest capacitance value for each dc voltage rating contained therein; for styles containing three dc voltage ratings, an equal number of sample units (58) shall be submitted representing the largest capacitance value for the two highest voltage ratings; the lowest voltage rating shall be represented by 115 sample units.

A.4. EXTENT OF QUALIFICATION

A.4.1 Single-type submission. Qualification within a style and dc voltage rating will be restricted to capacitance values equal to, or less than, those submitted. Qualification of the ±5 percent or closer capacitance tolerance automatically qualifies all other applicable capacitance tolerances.

A.4.2 Combined submission. Qualification of voltage ratings will be restricted to those submitted. Capacitance range and tolerances will be as specified in A.4.1.

A.5 SOLDER DIP (RETINNING) LEADS

A.5.1 Solder dip/retinning operations. Only the manufacturer (or his category B distributor) may solder dip/retin as follows.

a. As a corrective action if the lot fails the group A solderability test.

b. After the group A screening test. Following any solder dip or retinning process, the seal, insulation resistance, capacitance, and dissipation factor measurements shall be performed on 100 percent of the lot. The defective allowable (PDA) shall be the same as the group A, subgroup 1. Following these tests, the manufacturer shall submit the lot to the group A solderability test.

A.5.2 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 in accordance with MIL-STD-1276, the manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the capacitor. (NOTE: The 200 microinches maximum thickness requirement is not applicable.)

b. When the lead finish originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in A.5.2a, approval for the process to be used for solder dip shall be based on the following procedure:

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APPENDIX A

(1) Thirty samples of any capacitance value for each style and lead finish shall be subjected to the manufacturer's solder dip process. Following the solder dip process, the capacitors shall then be subjected to all group A, subgroup 1 post-electric tests, with no defects allowed.

(2) Ten of the 30 samples shall then be subjected to the solderability test, with no defects allowed.

(3) The remaining 20 samples shall be subjected to the resistance to soldering heat test, followed by the moisture resistance test (or hermetic seal test if the capacitor is hermetically sealed), with no defects allowed.

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DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2007-043)

Review activities:
Navy - AS, MC, OS, SH
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